

Phase 04 - NPG Annealings

Explanation and Operating Logic - Unified GUI

Release: 0.9.36 **Build:** 2026.08.11-r15

Operator and developer explanation of the current Phase 04 GUI, annealing recipe, parallel Keysight ramp-down, and finalization behavior.

1. Role in the synthesis workflow

- Run the final NPG oven annealing sequence after DP-DBBA evaporation.
- Accept the Phase 03 Keysight handoff and ramp the evaporator current down in parallel with the oven sequence.
- Monitor oven PID, CK-1 temperature, Keysight current and voltage.

2. Unified launcher and GUI

Unified launcher: START_NPG_CHAMBER.bat opens the main launcher. Each phase starts as a separate child process and owns its serial ports and hardware cleanup.

Phase-local shutdown: Abort / Safe Stop closes only the active phase after its safe hardware action. The unified launcher remains open and can be used again after the child process releases its ports.

Launcher close protection: while a phase is active, the launcher itself refuses to force-kill it. Use the phase GUI Abort / Safe Stop or let the phase finish normally.

GUI area	Purpose
Live plots	Oven temperature, CK-1 temperature, Keysight current and voltage.
Editable recipe	First/second stage targets and hold times can be changed through the GUI.
Phase sequence	Shows current stage and remaining time.
Abort / Safe Stop	Runs the Phase 04 abort hardware shutdown and closes only Phase 04.

3. Default annealing sequence

Stage	Default
Initial wait	200 C for 5 min
First stage	350 C, hold 15 min
Second stage	600 C, hold 40 min
Cooldown	Set PID to 0 C
Final hold	Keep PID at 0 C for 10 min, then finish

The current release intentionally has no final 30 C PID command. Normal completion leaves the oven PID setpoint at 0 C.

4. Keysight ramp-down

- The Phase 03 normal handoff supplies Phase 04 with Keysight OUTPUT ON at base current, normally 0.640 A.
- Phase 04 ramps current down independently of the oven recipe using its configured step and period.

- At electrical zero, the Keysight output is switched OFF.

5. Abort behavior

Abort / Safe Stop marks the phase as aborting, requests oven PID 0 C and forces the Keysight output OFF through the Phase 04 safety path. The phase process closes; the unified launcher remains open.

6. Data and operator checklist

- Verify the Phase 03 handoff state before starting Phase 04.
- Watch both oven PV/SV and Keysight current during the parallel sequence.
- Normal completion is time-defined: after the 0 C command, hold 10 min, save, then finish.
- Do not vent based only on software completion; wait for physically safe chamber/sample temperature.

Operational note: Software state is not a substitute for checking the real chamber hardware. Verify shutters, pressure, PID, Keysight output and any manually operated hardware before continuing.